

NON-LINEAR MODELS

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Non-linear Models

The linear regression model is given by

$$y_i = x_i' \theta + e_i. \quad (1)$$

But, in general a non-linear model is given by

$$y_i = m(x_i, \theta) + e_i. \quad (2)$$

In some cases we might use a parametric regression function $m(x_i, \theta) = E(y_i | x_i = x)$ which is a non-linear function of the parameters θ . We describe this setting as non-linear regression.

Nonlinear regression is sometimes adopted because the functional form $m(x_i, \theta)$ is suggested by an economic model.

Non-linear Models

Exponential Link Regression

Useful when it is desired to constrain the mean to be strictly positive.

$$m(x_i, \theta) = \exp(x_i' \theta). \quad (3)$$

Logistic Link Regression

This choice can be useful when the conditional mean is bounded between 0 and 1.

$$m(x_i, \theta) = (1 + \exp(-x_i' \theta))^{-1}. \quad (4)$$

Non-linear Models

Comes from Theory

CES production function

$$m(x_i, \theta) = \theta_0(\theta_1 K_i^{-\theta_2} + (1 - \theta_1)L_i^{-\theta_2})^{-\frac{\theta_3}{\theta_2}}, \quad (5)$$

where K is capital stock and L is labor, θ_1 and $1 - \theta_1$ determine the optimal distribution of inputs, θ_2 determines the (constant) elasticity of substitution, and θ_3 is the scale parameter.

If $\theta_2 = 1$ we have a linear or perfect substitutes function; If θ_2 approaches zero in the limit, we get the Cobb-Douglas production function; If θ_2 approaches negative infinity, we get the Leontief or perfect complements production function.

Non-linear Models

Comes from Theory

Sticky information **Phillips curve**

$$m(y_t, \pi_t, \theta) = \frac{\theta_0 \theta_1}{1 - \theta_1} y_t + \theta_1 \sum_{j=0}^k (1 - \theta_1)^j E_{t-1-j}(\pi_t + \theta_0 \Delta y_t), \quad (6)$$

where π_t is the inflation rate; y_t is the output gap; Δy_t is the growth rate of output gap; E_t is the conditional expectation; θ_0 and θ_1 are the real and information rigidities, respectively.

In this set-up, inflation depends on output, past expectations of current inflation, and past expectations of changes in current output growth



Threshold Model

Threshold regression

$$m(x_i, \theta) = \begin{cases} \theta_1 x_i & \text{if } q_i \leq \theta_3 \\ \theta_2 x_i & \text{if } q_i > \theta_3. \end{cases} \quad (7)$$

This model has been enormously influential in economics and is popular in current applied econometric practice. Some papers that use that model answer questions such as

- Threshold debt that slow down economic growth.
- Threshold inflation that slow down economic growth.
- Threshold currency mismatch that triggers the balance sheet effect.
- Threshold debt that constraint firm's investment.
- So one and so forth.

Threshold Model

The standard **threshold model**, introduced by [Tong \(1978\)](#), is given by

$$y_i = \begin{cases} \beta_1 x_i + e_i & \text{if } q_i \leq \gamma \\ \beta_2 x_i + e_i & \text{if } q_i > \gamma, \end{cases} \quad (8)$$

where,

- y is dependent variable and is a scalar.
- x is a set of regressors **assumed to be exogenous**.
- q is the threshold variable assumed to be exogenous **and must have a continuous distribution**.
- β_1 and β_2 are the slope parameters.
- γ is the threshold parameters.
- i indexes individuals or time.

Threshold Model

or in a compact form (8) can be written as

$$y_i = \beta_1 x_i 1(q_i \leq \gamma) + \beta_2 x_i 1(q_i > \gamma) + e_i, \quad (9)$$

where $1(q_i \leq \gamma)$ is an indicator variable that takes the value of 1 if $q_i \leq \gamma$ happens and 0 otherwise.

Features

- The model internally sorts the data on the basis of some threshold determinant into groups of observations.
- Each group of observations obeys the same linear model.
- We have one threshold parameter that divides the sample in two regimes.
- If $\beta_1 = \beta_2$ then the threshold model collapses to the linear model.

Asymptotic theory for least squares estimation (LSE) was developed by Chan (1993) and Hansen (2000).

Threshold Model

Estimation

Using a vector notation, the model can be written as

$$Y = X(\gamma)' \beta + e, \quad (10)$$

where,

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}, X(\gamma) = \begin{bmatrix} x_1 1(q_1 \leq \gamma) & x_1 1(q_1 > \gamma) \\ x_2 1(q_2 \leq \gamma) & x_2 1(q_2 > \gamma) \\ x_3 1(q_3 \leq \gamma) & x_3 1(q_3 > \gamma) \\ \vdots & \vdots \\ x_n 1(q_n \leq \gamma) & x_n 1(q_n > \gamma) \end{bmatrix}, e = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_n \end{bmatrix},$$

and $\beta = (\beta_1, \beta_2)'$

Threshold Model

Estimation

The parameters are estimated by least squares estimation by minimizing the sum of squared errors

$$S(\gamma, \beta) = e'e = (Y - X(\gamma)'\beta)'(Y - X(\gamma)'\beta), \quad (11)$$

Comments,

- The objective function, $S(\gamma, \beta)$, is smooth respect to the slope parameters, β , so we can take first order conditions.
- The objective function, $S(\gamma, \beta)$, is not smooth respect to the threshold parameters, γ so we cannot take first order conditions.
- And conventional gradient algorithms are not suitable for its maximization.

Threshold Model

Estimation

Using vector notation, standard estimation procedure follows the next steps

- 1 Form a grid on the empirical support of the threshold variable q_i .
- 2 For each value of γ on this grid, the conditional least square (CLS) estimator for β is

$$\hat{\beta}(\gamma) = (X(\gamma)'X(\gamma))^{-1}X(\gamma)'Y.$$

- 3 The vector of regression residuals is

$$\hat{e}(\gamma) = Y - X(\gamma)\hat{\beta}(\gamma).$$

- 4 The estimator of γ is the value that minimizes the sum of squared errors

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \hat{e}(\gamma)' \hat{e}(\gamma) \equiv \underset{\gamma}{\operatorname{argmin}} S(\gamma).$$

- 5 Once $\hat{\gamma}$ is obtained, the slope coefficient estimate is $\hat{\beta} = \hat{\beta}(\hat{\gamma})$.

Threshold Model

Estimation

Open the Matlab file ThresholdEstimation.m

We consider a model with only one exogenous variable. Then we use the following model to generate y_i ,

$$y_i = \beta_1 x_i 1(q_i \leq \gamma) + \beta_2 x_i 1(q_i > \gamma) + e_i, \quad (12)$$

We generate the threshold variable as $q_i \sim N(1/2, 1)$, the exogenous variable as $x_i \sim N(2, 3)$ and the error term as $e_{it} \sim N(0, 1)$.

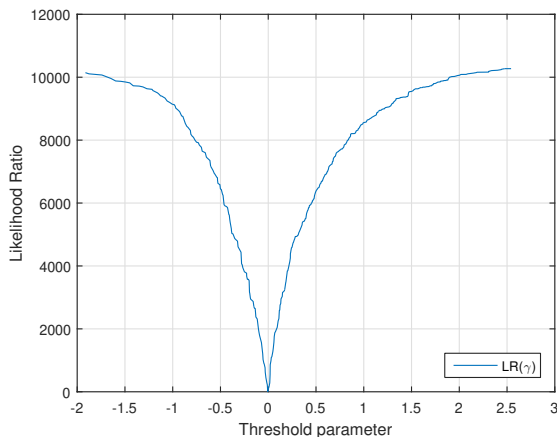
Consider $\beta_1 = 1$, $\beta_2 = -1$ and $\gamma = 0$.

Threshold Model

Estimation

The objective function looks like

Figure 1: Objective function in a threshold model



Threshold Model

Inference

Since the objective function, $S(\gamma, \beta)$, is smooth respect to the slope parameters, β , we can take the Taylor approximations and get asymptotic normality.

The objective function, $S(\gamma, \beta)$, is NOT smooth respect to the threshold parameter, γ , Hansen (2000) shows that the limiting distribution converges to a functional of a two-sided Brownian motion process.

Hansen (2000) argues the best ways to form confidence intervals for the threshold is to form the no-rejection region using the likelihood ratio statistic. To test hypothesis $H_0: \gamma = \gamma_0$, the likelihood ratio test is to reject large values of $LR(\gamma_0)$ where

$$LR(\gamma) = n \frac{S(\gamma) - S(\hat{\gamma})}{S(\hat{\gamma})}, \quad (13)$$

Threshold Model

Inference

Then, the asymptotic distribution of the likelihood ratio statistic is non-standard, yet free of nuisance parameters. Furthermore, the distribution function has the inverse

$$c(a) = -2 \ln \left(1 - \sqrt{1 - a} \right), \quad (14)$$

where a is the significance level. To form an asymptotic confidence interval for γ , the “no-rejection region” of confidence level $1 - a$ is the set of values of γ , such that $LR(\gamma) \leq c(a)$, where $LR(\gamma)$ is defined in (13) and $c(a)$ is defined in (14).

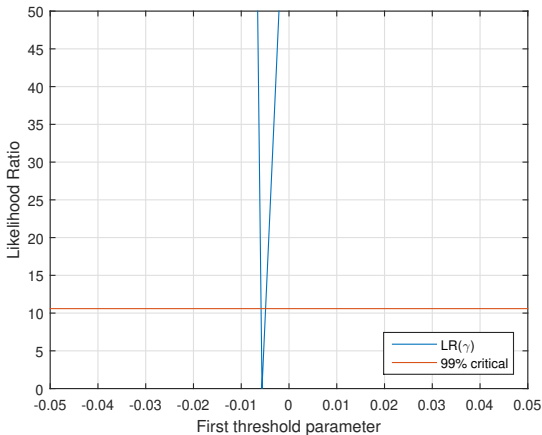
This is easiest to find by plotting $LR(\gamma)$ against γ and drawing a flat line at $c(a)$.

Threshold Model

Inference

Come back to the ThresholdEstimation.m program, the $LR(\gamma)$ statistics

Figure 2: Confidence interval construction for threshold



Threshold Model

Inference

The asymptotic 95% and 99% confidence intervals for the threshold are shown in the table below

Table 1: Asymptotic confidence interval in threshold model

	Threshold estimate (%)	95% confidence interval	99% confidence interval
$\hat{\gamma}$	0.0005	[-0.0025, 0.0005]	[-0.0025, 0.0006]

Note: Asymptotic critical values are reported in [Hansen \(2000\)](#).

Threshold Model

Testing

It is important to determine whether the threshold effect is statistically significant.

The hypothesis of no threshold effect in (1) can be represented by the linear constraint

$$H_0 : \beta_1 = \beta_2, \quad (15)$$

under H_0 the threshold model becomes the linear model, where the threshold γ is not identified, so classical tests have non-standard distributions.

[Hansen \(1996\)](#) shows that a bootstrap procedure are asymptotically valid. Thus, we are interested in reject the null hypothesis of no threshold.

Threshold Model

Testing

Compute $\hat{e}_i = y_i - \hat{\beta}_1 x_i 1(q_i \leq \hat{\gamma}) + \hat{\beta}_2 x_i 1(q_i > \hat{\gamma})$.

Generate the bootstrap dependent variable y_i^* from the distribution $N(0, \hat{e}_i^2)$.

Use the bootstrap sample to estimate the model under the null: $y_i = \beta_1 x_i + e_i$, and alternative: $y_i = \beta_1 x_i 1(q_i \leq \gamma) + \beta_2 x_i 1(q_i > \gamma) + e_i$, and calculate the bootstrap value of the likelihood ratio statistic $F = n \frac{S_0 - S(\hat{\gamma})}{S(\hat{\gamma})}$, where S_0 sum of squared errors of the model under the null.

Repeat this procedure a large number of times and calculate the percentage of draws for which the simulated statistic exceeds the actual.

The null of no threshold effect is rejected if the p -value is smaller than the desired critical value.

Application 1: Growth and Multiple Equilibria

Growth and Multiple Equilibria

Durlauf and Johnson (1995). suggest that cross-section growth behavior may be determined by initial conditions.

The model seeks to explain real GDP growth. The specification is

$$\begin{aligned} \ln Y_{i,85} - \ln Y_{i,60} = & \\ & (\beta_1 \ln Y_{i,60} + \beta_2 \ln I_{i,60} + \beta_3 \ln n_i + \beta_4 \ln S_i)1(Y_{i,60} \leq \gamma) \\ & (\beta_5 \ln Y_{i,60} + \beta_6 \ln I_{i,60} + \beta_7 \ln n_i + \beta_8 \ln S_i)1(Y_{i,60} > \gamma) \\ & + e_i, \end{aligned} \tag{16}$$

where

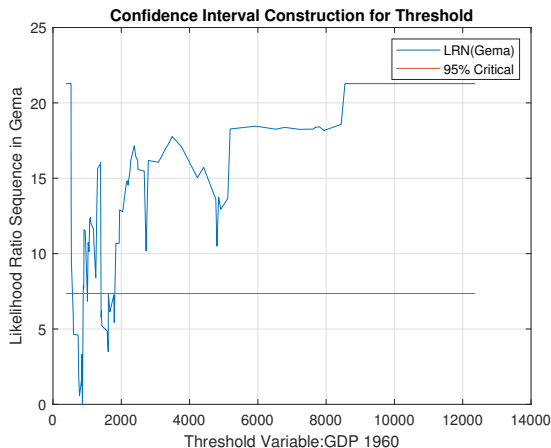
- Y is the real GDP per member of the population aged 15-64,
- I is the investment to GDP ratio,
- n is the growth rate of the working-age population,
- S is the fraction of working-age population enrolled in secondary school.

Application 1: Growth and Multiple Equilibria

Growth and Multiple Equilibria

The threshold estimate is 863 US dollars as it is shown in the figure below.

Figure 3: Confidence interval construction for threshold



Application 1: Growth and Multiple Equilibria

Growth and Multiple Equilibria

The asymptotic 95% confidence interval for the GDP per capita threshold is shown in the table below

Table 2: Asymptotic confidence interval in threshold model

	Threshold estimate (US\$)	95% confidence interval
$\hat{\gamma}$	863	[594, 1794]

Note: Asymptotic critical values are reported in [Hansen \(2000\)](#).

Application 1: Growth and Multiple Equilibria

Growth and Multiple Equilibria

Table 3: Slope estimates in each regime

Variable	Estimate	St. error
Regime1 GDP 1960 $\leq$ 863		
Constant	4.31	1.63
GDP 1960	-0.66	0.22
Investment	0.23	0.07
Pop Growth	-0.29	0.34
School	0.02	0.09
Regime2 GDP 1960 $>$ 863		
Constant	3.66	0.72
GDP 1960	-0.32	0.06
Investment	0.50	0.14
Pop Growth	-0.49	0.25
School	0.36	0.09

Note: Asymptotic critical values are reported in [Hansen \(2000\)](#).

Application 1: Growth and Multiple Equilibria

Growth and Multiple Equilibria

It is important to determine whether the threshold effect is statistically significant. For that we test null hypothesis of no threshold effect.

$$H_0 : \beta^{(1)} = \beta^{(2)}, \quad (17)$$

Hansen (1996) suggests to use a bootstrap procedure.

When GDP 1960 is used as a threshold variable, the bootstrap (using 1000 replications) p-value is 0.088.

When literacy rate is used as a threshold variable, the bootstrap (using 1000 replications) p-value is 0.2014.

Threshold Models in Panel Data

Hansen (1999) extended those threshold models to a panel threshold model as follows

$$y_{it} = \begin{cases} \mu_i + \beta_1 x_{it} + e_{it} & \text{if } q_{it} \leq \gamma \\ \mu_i + \beta_2 x_{it} + e_{it} & \text{if } q_{it} > \gamma \end{cases} \quad (18)$$

or in a compact form,

$$y_{it} = \mu_i + \beta_1 x_{it} 1(q_{it} \leq \gamma) + \beta_2 x_{it} 1(q_{it} > \gamma) + e_{it}, \quad (19)$$

Where

- q_{it} is the threshold variable assumed to be exogenous and must have a continuous distribution;
- x_{it} is a set of regressors assumed to be exogenous.
- μ_i is the individual fixed specific effect;
- γ is the threshold parameter;
- $\beta = (\beta_1, \beta_2)'$ are the slope parameters.

Threshold Models in Panel Data

Another compact representation of (18) is to set

$$x_{it}(\gamma) = \begin{pmatrix} x_{it}1(q_{it} \leq \gamma) \\ x_{it}1(q_{it} > \gamma) \end{pmatrix}$$

Then, equation (19) equals

$$y_{it} = \mu_i + \beta' x_{it}(\gamma) + e_{it}. \quad (20)$$

One traditional method to eliminate the individual fixed specific effect μ_i is to remove individual-specific means

$$\bar{y}_i = \mu_i + \beta' \bar{x}_i(\gamma) + \bar{e}_i, \quad (21)$$

Threshold Models in Panel Data

where $\bar{y}_i = T^{-1} \sum_{t=1}^T y_{it}$, $\bar{e}_i = T^{-1} \sum_{t=1}^T e_{it}$ and

$$\begin{aligned}\bar{x}_i(\gamma) &= \frac{1}{T} \sum_{t=1}^T x_{it}(\gamma) \\ &= \left(\begin{array}{c} \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{it} \leq \gamma) \\ \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{it} > \gamma) \end{array} \right)\end{aligned}$$

Taking the difference between (20) and (21) yields

$$y_{it}^* = \beta' x_{it}^*(\gamma) + e_{it}^*, \quad (22)$$

where $y_{it}^* = y_{it} - \bar{y}_i$, $x_{it}^*(\gamma) = x_{it}(\gamma) - \bar{x}_i(\gamma)$ and $e_{it}^* = e_{it} - \bar{e}_i$.

Threshold Models in Panel Data

Estimation

Using vector notation, standard estimation procedure follows the next steps

- 1 Form a grid on the empirical support of the threshold variable q_{it} .
- 2 For each value of γ on this grid, the conditional least square (CLS) estimator for β is

$$\hat{\beta}(\gamma) = (X^*(\gamma)'X^*(\gamma))^{-1}X^*(\gamma)'Y^*.$$

- 3 The vector of regression residuals is

$$\hat{e}^*(\gamma) = Y^* - X^*(\gamma)\hat{\beta}(\gamma).$$

- 4 The estimator of γ is the value that minimizes the sum of squared errors

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \hat{e}^*(\gamma)' \hat{e}^*(\gamma) \equiv \underset{\gamma}{\operatorname{argmin}} S(\gamma).$$

- 5 Once $\hat{\gamma}$ is obtained, the slope coefficient estimate is $\hat{\beta} = \hat{\beta}(\hat{\gamma})$.

Application 2: Balance Sheet and Currency Mismatch

- In conventional open economy models (Mundell-Fleming type), a depreciation of the local currency has a positive effect on the product.
- A significant portion of the debt held by Peruvian firms is denominated in US dollars (40 percent of liabilities are held in US dollar).
- Exchange rate depreciation increases the debt-to-asset ratio, making access to alternative sources of financing more difficult. Thus, these balance sheets negatively affect investment and production plans.
- If firms also hold assets and/or have income in US dollars that match their currency composition, then movements in the exchange rate should not affect their investment decisions.
- From which threshold currency mismatch the balance sheet effect dominates the competitiveness effect?

Application 2: Balance Sheet and Currency Mismatch

firm's investment activity depends on exchange rate movements

$$I_{it} = \beta q_t + \mu_i + \pi' z_{it} + u_{it}, \quad (23)$$

where I is the investment; q is the bilateral (Pen/USD) real exchange rate variation; μ_i is an unobserved firm specific-effect (assumed to be fixed as is common in empirical applications); z is a set of other determinants (controls) of investment; i refers to non-financial firm; and t refers time (year).

However, the impact depends on the relative strength of the competitiveness effect and the balance sheet effect. This effect can be decomposed as

$$\delta\beta = \delta + \theta D_{it-1}^* + \lambda X_{it}, \quad (24)$$

where D^* is the firm's liability denominated in foreign currency or US dollar debt and X is the firm's net export.

Application 2: Balance Sheet and Currency Mismatch

Plugging (24) into (23) produces

$$l_{it} = \theta(D_{it-1}^* \times q_t) + \lambda(X_{it} \times q_t) + \mu_i + \delta q_t + \pi' z_{it} + u_{it}. \quad (25)$$

Empirical analyses focused on the parameter θ , reveal that if this parameter is negative, evidence exists for the balance sheet effect.

Nonetheless, empirical studies found little evidence or no evidence on the negative significance of such parameter.

Application 2: Balance Sheet and Currency Mismatch

Ramírez-Rondán (2019) estimates the following threshold regression model

$$l_{it} = \theta_1(D_{it-1}^* \times q_t)1(CM_{it} \leq \gamma) + \theta_2(D_{it-1}^* \times q_t)1(CM_{it} > \gamma) + \mu_i + \delta q_t + \lambda(X_{it} \times q_t) + \pi'z_{it} + u_{it}, \quad (26)$$

where

- CM is the currency mismatch;
- D^* is the firm's liability denominated in foreign currency;
- q is the bilateral (Pen/USD) real exchange rate variation;
- θ_1 captures the negative effect of the balance sheet effect;
- θ_2 is the parameter in the second regime when firms have a low currency mismatch and then could hedge exchange rate risk;
- γ is a threshold currency mismatch level; and
- z is a set of other explanatory variables.

Application 2: Balance Sheet and Currency Mismatch

Table 4: Definitions

Variable	Definition
Investment	Investment is the expenditure in machinery and equipment net of fixed asset sales divided by total assets.
US dollar debt	Total liabilities in foreign currency expressed in terms of domestic currency as a percentage of total liabilities.
Currency mismatch	Total assets minus liabilities in US dollars expressed as a percentage of total assets in domestic currency.
Cash flow	Cash flow divided by total assets.
Working capital	Difference between current assets and current liabilities divided by total assets.
Firm size	Total sales in logarithm.
Leverage	Total liabilities divided by equity.
Net export	Difference between export and import at FOB prices divided by total assets.
Sales growth	Growth of total sales in real terms.
Real exchange rate	Growth of the bilateral rate Peruvian sol per US dollar divided by consumer price index (CPI).
Terms of trade	Terms of trade growth.
Dollarization ratio	Credit in US dollar expressed in terms of domestic currency divided by total credit of depositary corporations to the private sector.
Domestic interest rate	Lending interest rates of commercial banks in domestic currency (annual effective rates).

Note: All firm variables were deflated by the consumer price index.

Application 2: Balance Sheet and Currency Mismatch

Table 5: Tests for threshold effects

Test for single threshold	
F_1	15.1
P-value	0.07
(10%, 5%, 1% critical values)	(12.2, 18.2, 25.5)
Test for double threshold	
F_2	46.1
P-value	0.01
(10%, 5%, 1% critical values)	(20.3, 27.4, 40.0)
Test for triple threshold	
F_3	7.7
P-value	0.26
(10%, 5%, 1% critical values)	(10.9, 13.5, 19.5)

Application 2: Balance Sheet and Currency Mismatch

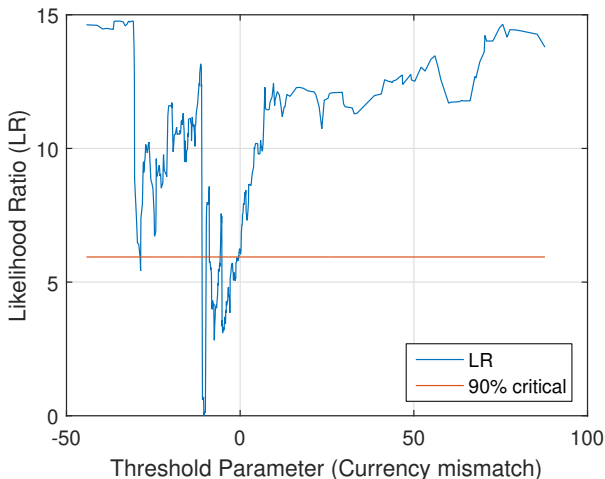
Table 6: Asymptotic confidence interval in threshold model

	Threshold estimate (%)	90% confidence interval	95% confidence interval	99% confidence interval
$\hat{\gamma}$	-10.4	[-10.9, -2.9]	[-28.8, 0.1]	[-30.4, 3.9]

Note: Asymptotic critical values are reported in Hansen (2000).

Application 2: Balance Sheet and Currency Mismatch

Figure 4: Confidence interval construction



Application 2: Balance Sheet and Currency Mismatch

Table 7: Percentage of firms in each regime by year

Firm class	2003	2004	2005	2006	2007	2008
Currency mismatch ≤ -10.4	40.5	41.9	36.5	24.3	28.4	36.5
Currency mismatch > -10.4	59.5	58.1	63.5	75.7	71.6	63.5
Firm class	2009	2010	2011	2012	2013	2014
Currency mismatch ≤ -10.4	28.4	24.3	29.7	33.8	33.8	29.7
Currency mismatch > -10.4	71.6	75.7	70.3	66.2	66.2	70.3

Application 2: Balance Sheet and Currency Mismatch

Table 8: Estimation results - Dependent variable: Investment

Explanatory variables	Linear model		Threshold model	
	Coefficient	LS SE	Coefficient	LS SE
Dollar debt $\times$ Real exchange rate (Currency mismatch ≤ -10.4)	–	–	-0.0042	0.0018
Dollar debt $\times$ Real exchange rate (Currency mismatch > -10.4)	–	–	0.0020	0.0014
Dollar debt $\times$ Real exchange rate	0.0000	0.0013	–	–
Real exchange rate	-0.0649	0.0827	-0.0414	0.0822
Net export $\times$ Real exchange rate	0.0005	0.0017	-0.0012	0.0017
Cash flow	0.0396	0.0193	0.0381	0.0191
Working capital	0.0149	0.0156	0.0183	0.0155
Firm size	-1.5085	0.5054	-1.4601	0.5013
Leverage	-0.0003	0.0032	-0.0009	0.0032
Sales growth	0.0043	0.0017	0.0043	0.0016
Terms of trade	0.0305	0.0145	0.0313	0.0144
Dollarization ratio	-0.0468	0.0173	-0.0487	0.0172
Domestic interest rate	-0.0134	0.0784	-0.0072	0.0778

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